NTIRE2026 – Robust Deepfake Detection Challenge

HIT VIRLAB
Jielun Peng Yabin Wang Yaqi Li Jincheng Liu Xiaopeng Hong*

March 17, 2026

1 Team details

a) Team name

HIT VIRLAB

b) Team leader name

Jielun Peng

c) Team leader institution and email (Please make sure is an active email)

`jielunpeng_hit@163.com`

d) Rest of the team members

Yabin Wang, Yaqi Li, Jincheng Liu, Xiaopeng Hong

e) Team website URL (if any)

No.

f) Affiliations

Harbin Institute of Technology

g) Usernames on the *NTIRE2026 – Robust Deepfake Detection Challenge* CodaBench leaderboard (development/validation and testing phases)

`hit_virlab`

h) Link to the codes/executables of the solution(s)

`https://drive.google.com/drive/folders/1E2u5w167EFp3jW3GrR4CntwmZ9vNnFzK?usp=drive_link`

1.1 Template

Please fill in the following template:

*Harbin Institute of Technology

HIT-VIRLAB

Title: Hierarchical Adaptive Feature Aggregation with Degraded-Original Consistency Learning for Robust Deepfake Detection
Members:
Jielun Peng(EMAIL), Yabin Wang, Yaqi Li, jincheng Liu, Xiaopeng Hong
Affiliations:
Harbin Institute of Technology

2 Contribution details

a) Title of the contribution

Hierarchical Adaptive Feature Aggregation with Degraded-Original Consistency Learning for Robust Deepfake Detection

b) A brief section describing the overall approach of your method. You can use team reports in Section 3 from this previous NTIRE challenge as an inspiration.

Our solution aims to improve the robustness of deepfake detection under strong perturbations and real-world degradations. Firstly, We construct a large-scale hybrid deepfake training dataset by collecting data from multiple publicly available sources, including FF++ [1], DFDC [2], FakeAVCeleb [3], Celeb-DF++ [4], DF40 [5], and DDL [6]. The resulting million-scale dataset contains diverse forgery generation methods and visual conditions, providing rich variations for training robust deepfake detectors.

The framework adopts a Vision Transformer (ViT) [7] backbone initialized with FSFM ViT-B [8] pretrained weights to extract features from input images. To better capture comprehensive forgery artifacts, we introduce a Hierarchical Adaptive Feature Aggregation (HAFA) module that leverages hierarchical features from multiple transformer layers. A learnable scoring network adaptively estimates token importance and aggregates informative tokens, enabling the model to integrate complementary cues ranging from low-level artifacts to high-level semantic information and emphasize critical forgery evidence.

Furthermore, we propose a Degraded-Original Consistency Learning (DOCL) strategy to improve robustness against common image degradations. During training, degraded samples are generated using a perturbation pipeline that simulates realistic distortions. A hierarchical consistency loss is applied to enforce feature consistency between original and degraded images. The final prediction is produced by an MLP classifier based on the concatenated aggregated hierarchical features.

c) References

[1] Rossler A, Cozzolino D, Verdoliva L, et al. Faceforensics++: Learning to detect manipulated facial images[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2019: 1-11.

[2] Dolhansky B, Bitton J, Pflaum B, et al. The deepfake detection challenge (dfdc) dataset[J]. arXiv preprint arXiv:2006.07397, 2020.

[3] Khalid H, Tariq S, Kim M, et al. FakeAVCeleb: A novel audio-video multimodal deepfake dataset[J]. arXiv preprint arXiv:2108.05080, 2021.

[4] Li Y, Zhu D, Cui X, et al. Celeb-df++: A large-scale challenging video deepfake benchmark for generalizable forensics[J]. arXiv preprint arXiv:2507.18015, 2025.

[5] Yan Z, Yao T, Chen S, et al. Df40: Toward next-generation deepfake detection[J]. Advances in Neural Information Processing Systems, 2024, 37: 29387-29434.

[6] Miao C, Zhang Y, Gao W, et al. DDL: A large-scale datasets for deepfake detection and localization in diversified real-world scenarios[J]. arXiv preprint arXiv:2506.23292, 2025.

[7] Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: Transformers for image recognition at scale[J]. arXiv preprint arXiv:2010.11929, 2020.

[8] Wang G, Lin F, Wu T, et al. Fsfm: A generalizable face security foundation model via self-supervised facial representation learning[C]//Proceedings of the Computer Vision and Pattern Recognition Conference. 2025: 24364-24376.

d) Representative image / diagram of the method

An overview of our proposed method is shown in Fig. 1.

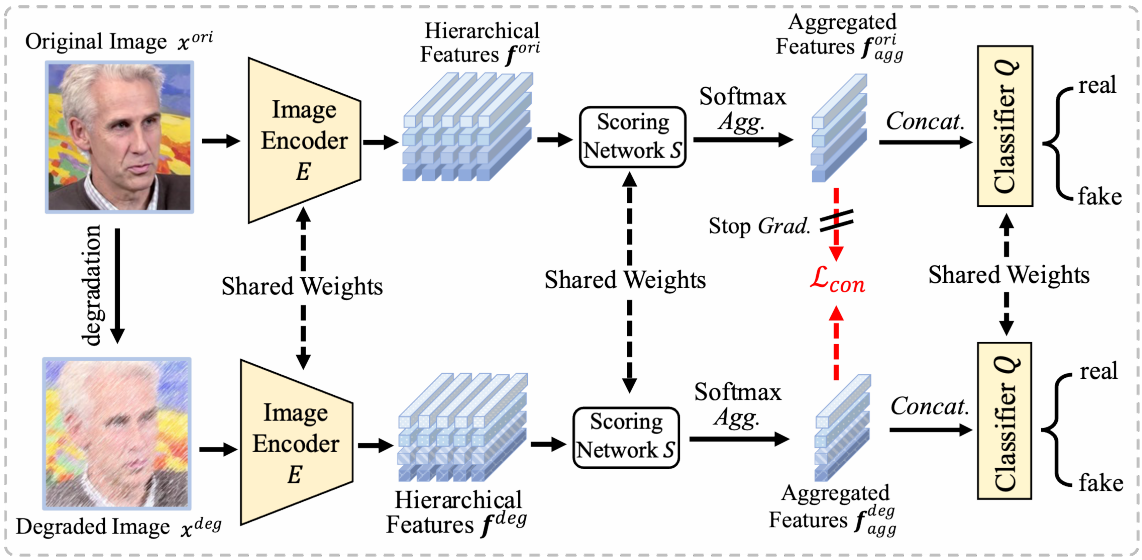

Figure 1: **Overview of our proposed method.** For each original image x^{ori}, a degraded counterpart x^{deg} is generated through a data perturbation pipeline. Both images are processed by a weight-shared Image Encoder E. The **Hierarchical Adaptive Feature Aggregation (HAFA) module** first extracts multi-level features from specific transformer layers and then employs Scoring Networks S to adaptively aggregate these hierarchical tokens into representative feature vectors $\boldsymbol{f}^{ori}_{agg}$ and $\boldsymbol{f}^{deg}_{agg}$. To enhance robustness, a **Degraded-Original Consistency Learning (DOCL) strategy** is applied via a hierarchical consistency loss $\mathcal{L}_{con}$ to align the aggregated features. Finally, these features are concatenated and fed into the Classifier Q to output the "real" or "fake" logits for original image x^{ori} and degraded image x^{deg}.

e) Have you tested previously published methods (yes/no)? If yes, please specify which methods and the results/problems you found.

No.

f) Other methods and baselines tested (even if results were not top competitive).

No.

3 Global Method Facts

a) Total method complexity: all stages

Model complexity: 92.864M parameters.

Average inference runtime: 0.146 s per image on a single NVIDIA L20 GPU.

b) Which pre-trained or external methods/models have been used (for any stage, if any)?

The image encoder of our model is initialized with pretrained weights from FSFM ViT-B [8].

[8] Wang G, Lin F, Wu T, et al. Fsfm: A generalizable face security foundation model via self-supervised facial representation learning[C]//Proceedings of the Computer Vision and Pattern Recognition Conference. 2025: 24364-24376.

c) Do you use extra data besides the competition data provided? (yes/no) If yes, please cite the datasets and briefly describe how you use them (pre-processing, number of images, etc.).

Yes, we collect data from multiple publicly available deepfake datasets, including FF++ [1], DFDC [2], FakeAVCeleb [3], Celeb-DF++ [4], DF40 [5], and DDL [6]. For datasets provided as images, we directly use the images for training. For video-based datasets, we uniformly sample frames from videos to obtain image data. After data collection, we construct a million-scale hybrid deepfake image dataset consisting of both real and manipulated samples.

[1] Rossler A, Cozzolino D, Verdoliva L, et al. Faceforensics++: Learning to detect manipulated facial images[C]//Proceedings of the IEEE/CVF international conference on computer vision. 2019: 1-11.

[2] Dolhansky B, Bitton J, Pflaum B, et al. The deepfake detection challenge (dfdc) dataset[J]. arXiv preprint arXiv:2006.07397, 2020.

[3] Khalid H, Tariq S, Kim M, et al. FakeAVCeleb: A novel audio-video multimodal deepfake dataset[J]. arXiv preprint arXiv:2108.05080, 2021.

[4] Li Y, Zhu D, Cui X, et al. Celeb-df++: A large-scale challenging video deepfake benchmark for generalizable forensics[J]. arXiv preprint arXiv:2507.18015, 2025.

[5] Yan Z, Yao T, Chen S, et al. Df40: Toward next-generation deepfake detection[J]. Advances in Neural Information Processing Systems, 2024, 37: 29387-29434.

[6] Miao C, Zhang Y, Gao W, et al. DDL: A large-scale datasets for deepfake detection and localization in diversified real-world scenarios[J]. arXiv preprint arXiv:2506.23292, 2025.

d) Training description

Image Encoder. We adopt a Vision Transformer (ViT) [7] architecture as the image encoder E, which consist of 12 Transformer layers. The image encoder E is initialized with pretrained weights from FSFM ViT-B [8], which is pre-trained using a self-supervised learning framework to learn fundamental and general facial representations from large-scale real face data. The learned representations capture intrinsic facial structures and provide a strong initialization for robust deepfake detection.

Hierarchical Adaptive Feature Aggregation. Existing deepfake detection methods typically utilize only the feature from the last layer for classification. However, the last layer mainly captures high-level semantic information, while fine-grained forgery artifacts often appear in earlier layers. As a result, relying solely on the last layer feature may overlook important low-level cues that are critical for deepfake detection.

To better exploit complementary information across feature hierarchies, we propose a Hierarchical Adaptive Feature Aggregation (HAFA) module that aggregates hierarchical features from multiple transformer layers.

Specifically, given an normalized image $x \in \mathbb{R}^{3\times224\times224}$, we adopt standard patch embeddings by dividing the image into 16×16 non-overlapping patches, flattening and linearly projecting each patch to obtain patch tokens $\boldsymbol{x} \in \mathbb{R}^{196\times768}$, where 196 denotes the number of patch tokens, and 768 is the embedding dimension of each token. The image encoder E then processes the patch tokens through a stack of transformer layers. Instead of using only the last layer feature, we extract features from layers 3, 6, 9, and 12, which capture complementary information ranging from low-level artifacts to high-level semantic cues. The extracted hierarchical features are denoted as $\boldsymbol{f} = \{\boldsymbol{f}_3, \boldsymbol{f}_6, \boldsymbol{f}_9, \boldsymbol{f}_{12}\}$, where $\boldsymbol{f}_l \in \mathbb{R}^{196\times768}$ denotes the feature extracted from the l-th layer of image encoder.

The extracted hierarchical features have a very high dimensionality, making it impractical to feed all tokens directly into the Deepfake classifier. In addition, different tokens contribute differently to deepfake detection. Therefore, a mechanism that can effectively aggregate the high-dimensional features while retaining the most critical information is essential.

Specifically, given the feature $\boldsymbol{f}_l \in \mathbb{R}^{196\times768}$ from $\boldsymbol{f}$, a learnable scoring network $S_l(\cdot)$, implemented as a two-layer MLP(Multilayer Perceptron) with layers mapping from the embedding dimension 768 to 128 and to 1, is employed to estimate token-level importance scores of each token $\{t_{l_i}\}_{i=1}^{196}$. These scores are normalized by a softmax function to obtain importance weights, then the aggregated feature $\boldsymbol{f}_{l,agg} \in \mathbb{R}^{768}$ is computed as the weighted sum of tokens:

$$\boldsymbol{f}_{l,agg} = \sum_{i=1}^{196} \frac{\exp\left(S_l(t_{l_i})\right)}{\sum_{j=1}^{196} \exp\left(S_l(t_{l_j})\right)} t_{l_i}. \tag{1}$$

Moreover, since tokens at different layers encode distinct structural and semantic information, we employ separate scoring networks $S_l(\cdot)$ for each layer, allowing for tailored importance evaluation according to the unique characteristics at each hierarchical level. Finally, we concatenate the aggregated features from different layers along the feature dimension to obtain the concatenated hierarchical aggregated feature $\boldsymbol{f}_{concat} \in \mathbb{R}^{3072}$.

Classifier Network. We employ a 3-layer MLP as the deepfake classifier Q, which takes as input the concatenated hierarchical aggregated feature $\boldsymbol{f}_{concat}$, outputs the logits, l, used to classify if a given sample is real or fake.

Degraded-Original Consistency Learning. In real-world scenarios, images distributed on the Internet inevitably undergo various degradations during the processes of compression, re-encoding, network transmission, and social media editing. These operations introduce diverse distortions that may significantly affect the visual characteristics of both real and fake images. In addition, malicious users may deliberately apply adversarial manipulations or post-processing operations to evade detection systems.

To improve the robustness of our model against both natural degradations and intentional attacks, we design a comprehensive data perturbation pipeline that simulates realistic degradations and apply it to both real and fake images. The specific perturbation methods used in this work are summarized in Tab. 1.

Table 1: Perturbation methods used in the proposed data perturbation pipeline.

Category	**Methods**
Geometric Transformations	Horizontal Flip, Image Translation, Image Resizing
Occlusion	Text Overlay
Noise	Gaussian Noise, Impulse Noise, Speckle Noise, Poisson Noise
Blur	Gaussian Blur, Lens Defocus Blur, Mean Blur
Color Distortion	Saturation Change, Color Shift, Grayscale, Quantization
Photometric Transformations	Brighten, Darken, Contrast Change
Compression	JPEG Compression

Based on the above perturbation pipeline, we further introduce a Degraded-Original Consistency Learning (DOCL) strategy. For each original image x^{ori}, we generate a degraded counterpart x^{deg} using the perturbation pipeline. x^{ori} and x^{deg} are separately fed into the encoder and the HAFA module to obtain hierarchical aggregated features $\boldsymbol{f}_{agg}^{ori} = \{\boldsymbol{f}_{3,agg}^{ori}, \boldsymbol{f}_{6,agg}^{ori}, \boldsymbol{f}_{9,agg}^{ori}, \boldsymbol{f}_{12,agg}^{ori}\}$ and $\boldsymbol{f}_{agg}^{deg} = \{\boldsymbol{f}_{3,agg}^{deg}, \boldsymbol{f}_{6,agg}^{deg}, \boldsymbol{f}_{9,agg}^{deg}, \boldsymbol{f}_{12,agg}^{deg}\}$.

To encourage the model to learn degradation-robust representations, we propose a hierarchical consistency loss $\mathcal{L}_{con}$ that enforces feature consistency between the original and degraded images at hierarchical levels:

$$\mathcal{L}_{con} = \sum_{l \in \mathcal{L}} \left(1 - \text{sim}\left(\text{sg}[\mathbf{f}_{l,agg}^{ori}], \mathbf{f}_{l,agg}^{deg}\right)\right), \quad \mathcal{L} = \{3, 6, 9, 12\} \tag{2}$$

where $\text{sim}(\cdot, \cdot)$ denotes the cosine similarity, and $\text{sg}[\cdot]$ represents the stop-gradient operation, meaning that gradients from $\mathcal{L}_{cons}$ are only propagated through the degraded features, while the features of the original image are treated as fixed targets.

Finally, the concatenated hierarchical aggregated features $\boldsymbol{f}_{concat}^{ori}$ and $\boldsymbol{f}_{concat}^{deg}$ are separately fed into the deepfake classifier Q to obtain the logits l^{ori} and l^{deg}.

Loss Function. To train the proposed framework, we optimize both the deepfake classification loss and the degraded-original hierarchical consistency loss. Specifically, The classification loss is formulated using the standard cross-entropy loss:

$$\mathcal{L}_{cls} = \mathcal{L}_{ce}(l^{ori}, y) + \mathcal{L}_{ce}(l^{deg}, y), \tag{3}$$

where l^{ori} and l^{deg} denote the logits predicted from the original and degraded images, respectively, y denotes the ground-truth label, and $\mathcal{L}_{ce}$ represents the cross-entropy loss.

In addition to the classification objective, the hierarchical consistency loss $\mathcal{L}_{con}$ is introduced to encourage the model to learn degradation-robust representations by aligning hierarchical features between original and degraded images. The overall training loss is defined as:

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda * \mathcal{L}_{con} \tag{4}$$

where λ=0.05 is a hyperparameter that controls the contribution of the consistency loss.

e) Testing description

During testing, each image is resized to 224×224 and normalized before being fed into the network. We apply test-time augmentation using horizontal flipping. In addition, Monte Carlo Dropout is used to perform multiple stochastic forward passes (10 times) for each input. The final deepfake probability is obtained by averaging the predictions from all augmented samples and stochastic runs.

f) Advantages of the proposed solution

The proposed solution is designed to improve the robustness of deepfake detection under strong perturbations and real-world degradations.

First, hierarchical features are exploited to capture forgery artifacts across multiple representation levels, ranging from low-level artifacts to high-level semantic cues. The Hierarchical Adaptive Feature Aggregation (HAFA) module then performs adaptive feature aggregation to highlight the informative tokens and emphasize critical forgery evidence. Finally, the degraded-original consistency learning (DOCL) strategy jointly optimizes the model using both original and degraded images, encouraging degradation-invariant representations by enforcing hierarchical consistency between them.

g) Results of the comparison to other approaches (if any)

No.

h) Results on other benchmarks (if any)

No.

i) Novelty degree of the solution, and if it has been previously published

The proposed solution is developed specifically for this challenge and has not been previously published. While it builds upon existing vision transformer architecture [7] and pretrained weights [8], we introduce the HAFA module for hierarchical adaptive feature aggregation and the DOCL strategy to improve robustness against real-world degradations.

[7] Dosovitskiy A, Beyer L, Kolesnikov A, et al. An image is worth 16x16 words: Transformers for image recognition at scale[J]. arXiv preprint arXiv:2010.11929, 2020.

[8] Wang G, Lin F, Wu T, et al. Fsfm: A generalizable face security foundation model via self-supervised facial representation learning[C]//Proceedings of the Computer Vision and Pattern Recognition Conference. 2025: 24364-24376.

j) **It is OK if the proposed solution is based on other works (papers, reports, Internet sources (links), etc). It is ethically wrong and misconduct if you do not properly give credit and hide this information.**

Please fill in Table 2 specifying the technical information (besides writing it), which should take 1 minute. Ensemble refers to model ensembles and self-ensembles.

Input	Training Time	Epochs	Extra data	Attention	Quantization	Base Model	# Params. (M)	Runtime	Ensembles	GPU
(224, 224, 3)	100h	20	Yes	Yes	No	Fsfm ViT-B [8]	92.864 Million	0.146 s / image	Yes	L20

Table 2: Overview of method specification.

4 Competition particularities

Any particularities of the solution for this competition in comparison to other challenges (if applicable).

The proposed solution is specifically designed to address the robustness requirement of the NTIRE 2026 Robust Deepfake Detection Challenge. The HAFA module exploits hierarchical features and adaptively aggregates informative tokens to better capture subtle forgery artifacts across multiple representation levels. In addition, the degraded-original consistency learning (DOCL) strategy simulates realistic perturbations and enforces feature consistency between original and degraded images, enabling the model to learn degradation-robust representations.

5 Technical details

Our method is implemented in Python using the PyTorch deep learning framework. The training data is constructed by combining several public deepfake datasets, including FF++ [1], DFDC [2], FakeAVCeleb [3], Celeb-DF++ [4], DF40 [5], and DDL [6]. The model is trained using the AdamW optimizer with a cosine annealing scheduler with warm restarts. A smaller learning rate of 1.0e-5 is applied to the pretrained image encoder, while the other modules are trained with a larger learning rate of 1.0e-4. Training is performed for 20 epochs with a total batch size of 512 on four NVIDIA L20 GPUs, which takes approximately 100 hours. The model is optimized using the cross-entropy loss and hierarchical consistency loss, as defined in Eq. 4. Mixed-precision training with PyTorch automatic mixed precision (AMP) and distributed data parallel training are adopted to improve computational efficiency.

6 Other details

a) Planned submission of a solution description paper at NTIRE 2026 workshop.

   No.

b) General comments and impressions of the *NTIRE2026 – Robust Deepfake Detection Challenge* (we appreciate your feedback to improve in future editions).

   The NTIRE2026 Robust Deepfake Detection Challenge has effectively emphasized the robustness of deepfake detection under distorted environments, which is crucial, as real-world images are often compressed and may be vulnerable to adversarial attacks. This focus on image manipulation is highly relevant to everyday scenarios. However, the challenge is currently limited to image-based detection, while in real life, we also encounter compression and manipulation in other modalities like audio and video. Expanding the challenge to include multi-modal data could better reflect the complexity of real-world deepfakes. Overall, the challenge has been a great experience, and we look forward to future challenges!

c) What do you expect from a new challenge in deepfake detection?

For future challenges, it would be great to focus on temporal aspects of deepfakes, such as detecting manipulations that occur over time rather than just in individual frames. Additionally, incorporating multi-modal deepfake detection, where both audio and video cues are considered, would reflect real-world scenarios better. Another interesting task would be detecting and localizing deepfakes in videos where only a small segment is manipulated, requiring models to pinpoint the altered section.